POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5021 | Semester V | Theory | 4 Credits

Industrial Management and Safety

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5021 as Industrial Management and Safety in Semester V for Mechanical Engineering. The detailed Revision 2026 official course page was unavailable. A publicly accessible SITTTR learning document titled Industrial Management & Safety and a public diploma syllabus were consulted as supporting material; neither is represented as the recovered official REV2026 detailed syllabus.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. Management and safety treatment must be reconciled with official REV2026 content when it becomes available; workplace decisions require applicable laws, employer procedures and competent authority.

Verified source note: Sources cover management, HR, quality/ISO/TQM, materials/inventory, CPM/PERT, quantitative methods, industrial safety and entrepreneurship as supporting study topics.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Industrial Management and Safety develops the managerial, quantitative and safety awareness needed to organise industrial work responsibly. It connects people, materials, quality, projects, decisions, risk control and entrepreneurship.

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Prerequisite foundation

Basic arithmetic, communication skills, engineering-workshop awareness and safety responsibility.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain management, organisational and human-resource principles.
2. Explain quality, materials and inventory-management methods.
3. Apply basic project-management and quantitative-decision concepts.
4. Apply industrial-safety risk controls and explain entrepreneurial planning awareness.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain management functions, organisational structures and HR practices.	Understand
CO2	Explain quality systems, purchasing, stores and inventory-control concepts.	Understand
CO3	Apply basic CPM/PERT and selected quantitative-management principles.	Apply
CO4	Apply accident-prevention and emergency-preparedness concepts to an industrial scenario.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Management, Organisation and Human Resources	Management functions, scientific/administrative approaches, ownership, structures, manpower planning, job analysis, training and incentives.	Approx. 15 hours	CO1	Module 1
2	Quality, Materials and Inventory	Quality planning, ISO/TQM concepts, audits, purchasing, stores, EOQ/ABC and sales awareness.	Approx. 15 hours	CO2	Module 2
3	Projects and Quantitative Decisions	Network planning, CPM/PERT, critical path/float, linear programming, transportation concepts and decision communication.	Approx. 16 hours	CO3	Module 3
4	Industrial Safety and Entrepreneurship	Safety terms/metrics, accident causation, hierarchy of controls, emergency response, hazardous work awareness and enterprise feasibility.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Management, Organisation and Human Resources

Management functions, scientific/administrative approaches, ownership, structures, manpower planning, job analysis, training and incentives.

CO1 · Approx. 15 hours

Quality, Materials and Inventory

Quality planning, ISO/TQM concepts, audits, purchasing, stores, EOQ/ABC and sales awareness.

CO2 · Approx. 15 hours

Projects and Quantitative Decisions

Network planning, CPM/PERT, critical path/float, linear programming, transportation concepts and decision communication.

CO3 · Approx. 16 hours

Industrial Safety and Entrepreneurship

Safety terms/metrics, accident causation, hierarchy of controls, emergency response, hazardous work awareness and enterprise feasibility.

CO4 · Approx. 14 hours

MODULE 1

Management, Organisation and Human Resources

Management functions, scientific/administrative approaches, ownership, structures, manpower planning, job analysis, training and incentives.

Approx. 15 hours **CO1** **Understand · Apply**

Learning goals

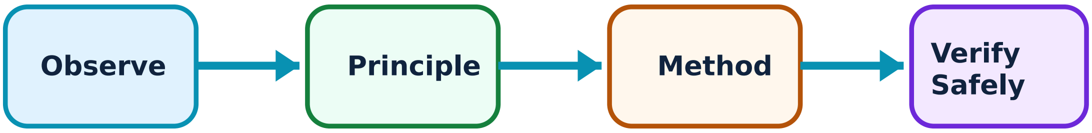
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Management, Organisation and Human Resources: identify the system, state its principle, follow the correct method and verify the result safely.

1. Management functions and organisation−

Planning, organising, staffing, directing and controlling coordinate resources toward objectives. Organisational structures allocate authority, communication and responsibility.

Study and application method

For a simple enterprise, identify objective, functions, reporting lines, decision rights and feedback/measurement method.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a simple enterprise, identify objective, functions, reporting lines, decision rights and feedback/measurement method.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not treat an organisation chart as a substitute for clear responsibility, communication or ethical conduct.

2. Human-resource planning−

HR planning aligns people, skills and work requirements. Job analysis, evaluation, selection, training, appraisal and fair incentives support capability and retention.

Study and application method

Write a basic job description with task, skill, responsibility, hazard and training requirement; distinguish job evaluation from performance appraisal.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write a basic job description with task, skill, responsibility, hazard and training requirement; distinguish job evaluation from performance appraisal.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Personnel decisions must be fair, evidence-based and compliant with applicable labour rules.

3. Wages and incentives−

Wage systems and incentives influence motivation, output, quality and safety. A good plan is understandable, fair, measurable and avoids encouraging unsafe/defective production.

Study and application method

State standard-time, actual-time, rate and quality/safety conditions before applying an approved incentive formula.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State standard-time, actual-time, rate and quality/safety conditions before applying an approved incentive formula.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never use incentives that reward bypassing safety or concealment of defects.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Quality, Materials and Inventory

Quality planning, ISO/TQM concepts, audits, purchasing, stores, EOQ/ABC and sales awareness.

Approx. 15 hours | CO2 | Understand · Apply

Learning goals

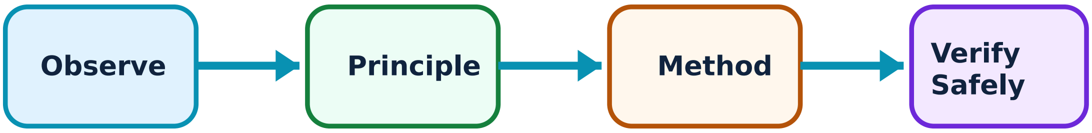
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Quality, Materials and Inventory: identify the system, state its principle, follow the correct method and verify the result safely.

1. Quality planning and TQM−

Quality planning translates customer/stakeholder needs into processes, standards, documentation and improvement. TQM promotes organisation-wide, evidence-based continuous improvement.

Study and application method

Define requirement, process measure, target, evidence source, corrective action and review cycle for a simple process.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define requirement, process measure, target, evidence source, corrective action and review cycle for a simple process.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Certification or documentation alone does not demonstrate real quality performance.

2. Purchasing and stores−

Purchasing secures the right material, quality, quantity, source, time and cost. Stores receive, identify, preserve, issue and record inventory to support production without excess loss.

Study and application method

Trace requisition → quotation/source evaluation → purchase order → receipt/inspection → storage → issue → records.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Do not substitute unapproved material or ignore traceability/inspection requirements.

3. Inventory control−

EOQ balances ordering and holding costs under stated assumptions; ABC analysis focuses control on high-value items. Reorder levels depend on demand, lead time and safety stock.

Study and application method

List demand, ordering/holding costs, lead time and service-risk assumptions; calculate only with consistent units and stated formula.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List demand, ordering/holding costs, lead time and service-risk assumptions; calculate only with consistent units and stated formula.

> **Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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> **Common mistake / precaution:** EOQ is a model, not a universal answer—uncertainty, obsolescence and supplier reliability matter.

> **Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Projects and Quantitative Decisions

Network planning, CPM/PERT, critical path/float, linear programming, transportation concepts and decision communication.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

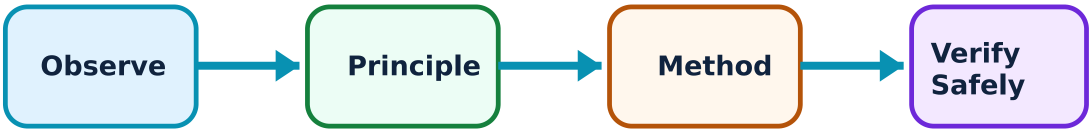
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Projects and Quantitative Decisions: identify the system, state its principle, follow the correct method and verify the result safely.

1. Network planning−

CPM/PERT represent activities and precedence to estimate project duration, float and critical activities. They make dependencies visible and support monitoring.

Study and application method

Define activities, predecessor logic, durations and nodes; calculate/identify critical path and explain effect of delay on critical versus noncritical activity.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define activities, predecessor logic, durations and nodes; calculate/identify critical path and explain effect of delay on critical versus noncritical activity.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A network is only useful when logic and duration data are realistic and regularly updated.

2. CPM and PERT−

CPM typically uses known durations; PERT uses optimistic, most-likely and pessimistic estimates to represent uncertainty. Both support planning, not certainty.

Study and application method

Use the prescribed formula/method, label assumptions and report expected time/critical path/float with units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use the prescribed formula/method, label assumptions and report expected time/critical path/float with units.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not present estimated completion dates as guarantees without risk and resource review.

3. Quantitative methods−

Linear programming and transportation models support allocation decisions when objectives, constraints and variables are specified. Results require practical validation.

Study and application method

Define decision variables, objective, constraints and non-negativity condition before drawing/solving an approved small graphical model.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define decision variables, objective, constraints and non-negativity condition before drawing/solving an approved small graphical model.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Mathematical optimum may be infeasible if capacity, safety, quality or human constraints are missing.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Industrial Safety and Entrepreneurship

Safety terms/metrics, accident causation, hierarchy of controls, emergency response, hazardous work awareness and enterprise feasibility.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

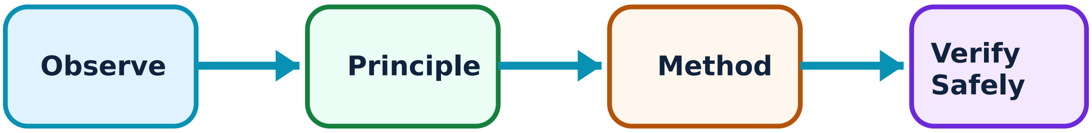
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Industrial Safety and Entrepreneurship: identify the system, state its principle, follow the correct method and verify the result safely.

1. Accident prevention and safety systems−

Accidents arise from interacting technical, environmental, organisational and human factors. Safety systems identify hazards, assess risk, control exposure, inspect performance and learn from incidents.

Study and application method

Describe task, hazard, exposed person, consequence and controls using elimination, substitution, engineering, administration and PPE hierarchy.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe task, hazard, exposed person, consequence and controls using elimination, substitution, engineering, administration and PPE hierarchy.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: PPE is not the first control for a serious hazard; do not start work when critical controls are absent.

2. Emergency preparedness−

Preparedness defines alarms, communication, evacuation, first aid, isolation and competent response for foreseeable emergencies such as fire, electrical shock or chemical release.

Study and application method

For a scenario, state trigger, immediate safe action, notification route, evacuation/muster and authorised responder role.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a scenario, state trigger, immediate safe action, notification route, evacuation/muster and authorised responder role.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Do not attempt rescue beyond training/equipment; call appropriate emergency support.

3. Entrepreneurship and feasibility−

Entrepreneurship turns an opportunity into a viable enterprise through customer understanding, resources, risk management, finance, operations and ethical compliance. A feasibility report tests assumptions.

Study and application method

Outline problem/customer, value proposition, resources, costs, risks, compliance and measurable milestones for a small concept.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Outline problem/customer, value proposition, resources, costs, risks, compliance and measurable milestones for a small concept.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Do not make financial/legal commitments from a classroom feasibility exercise without professional advice.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Management, Organisation and Human Resources

Use the concept flow to revise observation, principle, method and verification.

Module 2: Quality, Materials and Inventory

Use the concept flow to revise observation, principle, method and verification.

Module 3: Projects and Quantitative Decisions

Use the concept flow to revise observation, principle, method and verification.

Module 4: Industrial Safety and Entrepreneurship

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare an organisational chart and training matrix for a small workshop.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Perform an ABC classification or EOQ example with stated assumptions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create a small CPM/PERT network from supplied activities.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Complete a risk assessment and emergency-response outline for a simulated industrial task.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Management, Organisation and Human Resources

Explain Management functions and organisation and state one correct method, application or precaution.−

Planning, organising, staffing, directing and controlling coordinate resources toward objectives. Organisational structures allocate authority, communication and responsibility.

Answer framework: For a simple enterprise, identify objective, functions, reporting lines, decision rights and feedback/measurement method.

Important point: Do not treat an organisation chart as a substitute for clear responsibility, communication or ethical conduct.

Explain Human-resource planning and state one correct method, application or precaution.−

HR planning aligns people, skills and work requirements. Job analysis, evaluation, selection, training, appraisal and fair incentives support capability and retention.

Answer framework: Write a basic job description with task, skill, responsibility, hazard and training requirement; distinguish job evaluation from performance appraisal.

Important point: Personnel decisions must be fair, evidence-based and compliant with applicable labour rules.

Explain Wages and incentives and state one correct method, application or precaution.−

Wage systems and incentives influence motivation, output, quality and safety. A good plan is understandable, fair, measurable and avoids encouraging unsafe/defective production.

Answer framework: State standard-time, actual-time, rate and quality/safety conditions before applying an approved incentive formula.

Important point: Never use incentives that reward bypassing safety or concealment of defects.

Module 2 — Quality, Materials and Inventory

Explain Quality planning and TQM and state one correct method, application or precaution.−

Quality planning translates customer/stakeholder needs into processes, standards, documentation and improvement. TQM promotes organisation-wide, evidence-based continuous improvement.

Answer framework: Define requirement, process measure, target, evidence source, corrective action and review cycle for a simple process.

Important point: Certification or documentation alone does not demonstrate real quality performance.

Explain Purchasing and stores and state one correct method, application or precaution.−

Purchasing secures the right material, quality, quantity, source, time and cost. Stores receive, identify, preserve, issue and record inventory to support production without excess loss.

Answer framework: Trace requisition → quotation/source evaluation → purchase order → receipt/inspection → storage → issue → records.

Important point: Do not substitute unapproved material or ignore traceability/inspection requirements.

Explain Inventory control and state one correct method, application or precaution.−

EOQ balances ordering and holding costs under stated assumptions; ABC analysis focuses control on high-value items. Reorder levels depend on demand, lead time and safety stock.

Answer framework: List demand, ordering/holding costs, lead time and service-risk assumptions; calculate only with consistent units and stated formula.

Important point: EOQ is a model, not a universal answer—uncertainty, obsolescence and supplier reliability matter.

Module 3 — Projects and Quantitative Decisions

Explain Network planning and state one correct method, application or precaution.−

CPM/PERT represent activities and precedence to estimate project duration, float and critical activities. They make dependencies visible and support monitoring.

Answer framework: Define activities, predecessor logic, durations and nodes; calculate/identify critical path and explain effect of delay on critical versus noncritical activity.

Important point: A network is only useful when logic and duration data are realistic and regularly updated.

Explain CPM and PERT and state one correct method, application or precaution.−

CPM typically uses known durations; PERT uses optimistic, most-likely and pessimistic estimates to represent uncertainty. Both support planning, not certainty.

Answer framework: Use the prescribed formula/method, label assumptions and report expected time/critical path/float with units.

Important point: Do not present estimated completion dates as guarantees without risk and resource review.

Explain Quantitative methods and state one correct method, application or precaution.−

Linear programming and transportation models support allocation decisions when objectives, constraints and variables are specified. Results require practical validation.

Answer framework: Define decision variables, objective, constraints and non-negativity condition before drawing/solving an approved small graphical model.

Important point: Mathematical optimum may be infeasible if capacity, safety, quality or human constraints are missing.

Module 4 — Industrial Safety and Entrepreneurship

Explain Accident prevention and safety systems and state one correct method, application or precaution.−

Accidents arise from interacting technical, environmental, organisational and human factors. Safety systems identify hazards, assess risk, control exposure, inspect performance and learn from incidents.

Answer framework: Describe task, hazard, exposed person, consequence and controls using elimination, substitution, engineering, administration and PPE hierarchy.

Important point: PPE is not the first control for a serious hazard; do not start work when critical controls are absent.

Explain Emergency preparedness and state one correct method, application or precaution.−

Preparedness defines alarms, communication, evacuation, first aid, isolation and competent response for foreseeable emergencies such as fire, electrical shock or chemical release.

Answer framework: For a scenario, state trigger, immediate safe action, notification route, evacuation/muster and authorised responder role.

Important point: Do not attempt rescue beyond training/equipment; call appropriate emergency support.

Explain Entrepreneurship and feasibility and state one correct method, application or precaution.−

Entrepreneurship turns an opportunity into a viable enterprise through customer understanding, resources, risk management, finance, operations and ethical compliance. A feasibility report tests assumptions.

Answer framework: Outline problem/customer, value proposition, resources, costs, risks, compliance and measurable milestones for a small concept.

Important point: Do not make financial/legal commitments from a classroom feasibility exercise without professional advice.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Management, Organisation and Human Resources.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Quality, Materials and Inventory.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Projects and Quantitative Decisions.

6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Industrial Safety and Entrepreneurship.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Management functions, scientific/administrative approaches, ownership, structures, manpower planning, job analysis, training and incentives..
2. Develop a complete answer or practical plan for: Quality planning, ISO/TQM concepts, audits, purchasing, stores, EOQ/ABC and sales awareness..
3. Develop a complete answer or practical plan for: Network planning, CPM/PERT, critical path/float, linear programming, transportation concepts and decision communication..
4. Develop a complete answer or practical plan for: Safety terms/metrics, accident causation, hierarchy of controls, emergency response, hazardous work awareness and enterprise feasibility..

LAST-MILE REVISION

Quick Revision

Module 1: Management, Organisation and Human Resources

Management functions, scientific/administrative approaches, ownership, structures, manpower planning, job analysis, training and incentives.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Quality, Materials and Inventory

Quality planning, ISO/TQM concepts, audits, purchasing, stores, EOQ/ABC and sales awareness.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Projects and Quantitative Decisions

Network planning, CPM/PERT, critical path/float, linear programming, transportation concepts and decision communication.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Industrial Safety and Entrepreneurship

Safety terms/metrics, accident causation, hierarchy of controls, emergency response, hazardous work awareness and enterprise feasibility.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Management functions and organisation	Planning, organising, staffing, directing and controlling coordinate resources toward objectives.
Human-resource planning	HR planning aligns people, skills and work requirements.
Wages and incentives	Wage systems and incentives influence motivation, output, quality and safety.
Quality planning and TQM	Quality planning translates customer/stakeholder needs into processes, standards, documentation and improvement.
Purchasing and stores	Purchasing secures the right material, quality, quantity, source, time and cost.
Inventory control	EOQ balances ordering and holding costs under stated assumptions; ABC analysis focuses control on high-value items.
Network planning	CPM/PERT represent activities and precedence to estimate project duration, float and critical activities.
CPM and PERT	CPM typically uses known durations; PERT uses optimistic, most-likely and pessimistic estimates to represent uncertainty.
Quantitative methods	Linear programming and transportation models support allocation decisions when objectives, constraints and variables are specified.
Accident prevention and safety systems	Accidents arise from interacting technical, environmental, organisational and human factors.
Emergency preparedness	Preparedness defines alarms, communication, evacuation, first aid, isolation and competent response for foreseeable emergencies such as fire, electrical shock or chemical release.
Entrepreneurship and feasibility	Entrepreneurship turns an opportunity into a viable enterprise through customer understanding, resources, risk management, finance, operations and ethical compliance.

LEARNING RESOURCES

References

Recommended industrial-management and safety references

1. O. P. Khanna, Industrial Engineering and Management.
2. L. S. Srinath, PERT and CPM: Principles and Applications.
3. L. M. Deshmukh, Industrial Safety Management.

Approved supporting syllabus sources

- https://www.sitttrkerala.ac.in/index.php?r=site%2Fdownload-file&id=16
- https://gwpctsr.ac.in/wp-content/uploads/ee5-syllabus.pdf

These sources support the study handbook while the official Course 5021 REV2026 detail is unavailable; they do not replace legal, safety or organisational requirements.